



POLATIS® Series 6000 Multimode Optical Switch User Manual

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Document conventions

 **Tip:** A useful suggestion to help the user get the most out of the product.

 **Note:** Something a user should be aware of and take into account.

 **Warning:** Something that risks physical injury, damage to the switch, or irretrievable loss of data.

1 HUBER+SUHNER Polatis software rights

HUBER+SUHNER Polatis own the rights to the software embedded within the unit described in this manual*. This software is supplied solely for operation of the unit as described in this manual. Neither the embedded software nor any part thereof may be extracted, modified, reverse compiled, reproduced, distributed, or used for any other purpose than intended.

The specifications and information regarding these products are subject to change without notice. All statements, information, and recommendations in this manual are believed to be accurate, but are presented without warranty of any kind, express or implied.

* Certain components of the software are provided to HUBER + SUHNER Polatis under license. The full licensing agreements are available from Customer Support or can be downloaded online from www.polatis.com/support.

2 About this manual

This user manual is written to be used with the following existing HUBER+SUHNER Polatis documentation. Please reference the latest version of these documents available on the HUBER+SUHNER Polatis Support Website at www.polatis.com/support for the proper installation, quick configuration, safe operation and user interface specifications of all HUBER+SUHNER Polatis optical switching products.

1. Polatis Quick Start Guide
 - a. 7001-011-XX_Polatis_Quick_Start_Guide.pdf
2. Polatis User Interface Operations Manuals
 - a. Web GUI: 7001-007-XX H+S_Polatis_Web_GUI_User_Manual
 - b. TL1 Command Line Interface: 7001-009-XX H+S_Polatis_TL1_User_Manual
 - c. SNMP Network Protocol: 7001-023-XX SNMP_User_Guide
 - d. SCPI Command Line Interface: 7001-008-XX Polatis_SCPI_User_Manual
 - e. NETCONF: 7001-012-XX H+S_Polatis_NETCONF_User_Manual
 - f. RESTCONF: 7001-006-XX H+S_Polatis_RESTCONF_User_Manual

The Series 6000 Multimode user manual references the above and covers those features that are unique to the Series 6000 Multimode. The switch is configured to be plug and play, and as such requires only minimal configuration for use in a system. All required settings can be accessed via the built-in Web GUI described in this manual. Not all commands listed in the referenced manuals are supported in the Polatis Series 6000 Multimode switch.

3 Overview

POLATIS optical switches are electrically controlled fiber optic switches. Optical fibers are connected to input and output ports and the switch internally connects the optical paths as requested by external commands.

This manual covers the installation and commissioning of the POLATIS Series 6000 Multimode optical switch. This multimode fiber switch features up to 16 input ports and 16 output ports.

3.1 Features

Suitable for a wide range of applications, POLATIS optical switches offer ultra-low optical loss on all paths. Electrical connectivity to the switch is provided via serial and Ethernet connections. The switch is controlled using a web-based Graphical User Interface (GUI) or text-based Command Line Interface (CLI) over serial or Ethernet connections.

The operation of the switch is based on piezo-electric actuation and position sensing. This achieves ultra-low insertion loss and polarization dependent loss (PDL) by simply steering the input optical beams to point directly at the required output position.

Key Features

- Ultra-low insertion loss
- Low polarization dependent loss
- Fast switching speed
- Excellent repeatability
- High optical power handling
- Operates with dark or lit fiber
- Non-blocking architecture
- Bi-directional operation
- Protocol and bit rate independent

3.2 Description

The POLATIS Series 6000 Multimode switch is available in matrix sizes up to 16x16 ports in a 1RU configuration with up to 32 LC Multimode optical connectors mounted on the front panel as shown below. It is also available with other optical connector configurations in 2RU and 3RU chassis heights. **The switch must be installed on a solid surface capable of supporting 15 kg (33 pounds) and cannot be suspended from only the front panel mounting ears.**

Front View



Rear View



3.3 Model number

The Series 6000 Multimode switch is identified by a model number similar to the following: Polatis, S-VST-16x16-LP8-HMHNS. The different components of the model number are defined below, where each highlighted variable has a description directly beneath it. Not all variables are available in both grades.

X - _ _ _ - _ _ _ X _ _ _ - _ _ _ - _ _ _ - _ _ _

I = Instrumentation Grade

S = Secure Grade

_ - **OST** - _ _ _ X _ _ _ - _ _ _ - _ _ _ - _ _ _

OST = Optical Switch Tray

_ - _ _ _ - **XXX** X _ _ _ - _ _ _ - _ _ _ - _ _ _

004 to 016 input ports

_ - _ _ _ - _ _ _ X **XXX** - _ _ _ - _ _ _ - _ _ _

004 to 016 output ports

_ - _ _ _ - _ _ _ X _ _ _ - **X** _ _ _ - _ _ _ - _ _ _

Connector Type -

L = LC

H = LC-HD

M = MTP-8

N = MTP-12

F = FC

C = SC

E = Diamond E2000

D = Diamond E3000

X = no connector

_ - _ _ _ - _ _ _ X _ _ _ - **X** _ _ _ - _ _ _ - _ _ _

Polish -

P = Flat Polish Multimode

A = APC Duplex

V = PC Simplex

B = APC Simplex

S = PC QFSP Duplex

Q = APC QFSP Duplex

_ - _ _ _ - _ _ _ X _ _ _ - **X** _ _ _ - _ _ _ - _ _ _

8 = Multimode 50/125µm graded index 0.20 NA

9 = Multimode 62.5/125µm graded index 0.27NA

_ - _ _ _ - _ _ _ X _ _ _ - _ _ _ - **X** _ _ _ - _ _ _

Interface -

D = Dual Ethernet NIC & serial

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H = Hot swap NICs

__ - __ - __ - __ - __ X __ - __ - __ - __ - __ M __ - __ - __ - __
Protocol -

M = Multiprotocol (includes SCPI, TL1, SNMP, NETCONF and RESTCONF)

__ - __ - __ - __ - __ X __ - __ - __ - __ - __ X __ - __ - __ - __
Power –Mains Connector Type

H = Dual Hot Swap AC mains (autosensing 100-240VAC)

T = Twin -48VDC hot swap (nominally -48VDC)

__ - __ - __ - __ - __ X __ - __ - __ - __ - __ N __ - __ - __ - __
Environmental –

N = Normal

__ - __ - __ - __ - __ X __ - __ - __ - __ - __ X __ - __ - __ - __
Customization –

S for Standard

V for Variant Code

__ - __ - __ - __ - __ X __ - __ - __ - __ - __ V- XXX

Optional Variant Code only if V specified above -

001 to 099

4 Hazards and cautions

4.1 Child handling

This equipment is not suitable for use in locations where children are likely to be present.

4.2 Electrical safety

- **Field Serviceable parts:** The optical connectors (see warning on laser eye safety), hot-swap power supply units (PSUs), Network Interface Controller(s) (NICs) and fans are the only field-serviceable parts.

Warning: Do not attempt to disassemble or access any parts internal to the chassis tray, the optical control system contains high voltages and has no user-serviceable parts inside. Trained personnel only should be allowed access to the rear of the switch.

- **Multiple power sources:** This product contains dual hot-swap power feeds. The rear of the tray has a yellow warning label that contains the following symbols and statement:



AC Power label example:



Warning: The product will remain powered unless both power feeds are disconnected from the switch, or the power switch is in the OFF position (marked 0 on the switch). Make sure both power feeds are disconnected before servicing of the product. Servicing of the switch is only permitted by trained personnel.

- **DC power option: Nominal -48VDC:** A Series 6000 Multimode switch can be powered by up a nominal -48VDC power source, if fitted with the appropriate dual hot-swap power supplies (dependent on customer option at time of order). See [Section 6.4.2](#) for how to safely connect to this power source type.

The following warnings must be adhered to when using a DC power source:

- **Warning:** An over current breaker must be provided, between the DC power source and the product. This can be one for both power inlets or one for each inlet. This needs to be rated to trip at a maximum of 10A.
- **Warning:** A service disconnect must be provided for the equipment, this must be marked that it is for the equipment and displayed clearly nearby the product.
- **Warning:** The product must be provided with a Ground (Earth) connection, either at the supply terminal strip or via the Ground (Earth) bonding stud on the chassis (located between the power switches).

4.3 Rack system instructions

The following rack-mount instructions should be observed during installation of the tray. Additional information is provided in [Section 6 Mounting and installation](#).

- **Elevated operating ambient temperature:** If installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient. Therefore, consideration should be given to installing the equipment in an environment compatible with the maximum ambient temperature (T_{ma}) specified by the manufacturer.
- **Reduced air flow:** Installation of the equipment in a rack should be such that the amount of air flow required for safe operation of the equipment is not compromised.
- **Mechanical loading:** Mounting of the equipment in the rack should be such that a hazardous condition is not achieved due to uneven mechanical loading.
- **Circuit overloading:** Consideration should be given to the connection of the equipment to the supply circuit and the effect that overloading of the circuits might have on overcurrent protection and supply wiring. Appropriate consideration of equipment nameplate ratings should be used when addressing this concern.
- **Reliable grounding (earthing):** Reliable grounding (earthing) of rack-mounted equipment should be maintained. Particular attention should be given to supply connections other than direct connections to the branch circuit (for example, use of power strips).

4.4 Laser eye safety

Polatis switches do not generate laser signals internally, but the switch is capable of passing through high optical intensities with minimal optical loss.

Within the limits of the maximum optical input power specified under [Absolute maximum ratings](#) below, the equipment connected to the Optical Switch Module (OSM) partially determines the level of safety hazard.

It is the responsibility of the user to assess what level of optical intensity may exist at input and output connectors and to label the system with the appropriate laser safety classification.

4.5 Absolute maximum ratings

4.5.1 Electrical

The absolute maximum electrical ratings of the POLATIS Series 6000 Multimode optical switch are shown in the table below:

Table 1 Absolute maximum electrical ratings

Power supply designation	H	T
Supply voltage	100-240 VAC (47-63Hz) 1 A	-48 VDC Nominal -40 to -72 VDC 1 A

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Maximum electrical power dissipation	28 W	28 W
--------------------------------------	------	------

 **Note:** This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

4.5.2 Optical

The maximum optical input power is common across the POLATIS optical switch product family.

Table 2 Max optical input power

Max optical input power	+27 dBm per input
-------------------------	-------------------

Refer also to the [Connecting and Disconnecting](#), [Fiber connection](#) sections in the [Maintenance and Operation](#) chapter.

5 Dimensions and weight

The Polatis Series 6000 Multimode 1RU switch configuration is shown below with dimensions in millimeters: 550 deep x 440 wide x 44mm (1RU) tall, with front mounting ears 479 mm in width. (21.7 x 17.3 x 1.75 inches with front mounting ears 19" in width). The switch weighs 11kg (24.3 pounds). **The switch must be installed on a solid surface capable of supporting 15kg (33 pounds) and cannot be suspended from only the front panel mounting ears.**

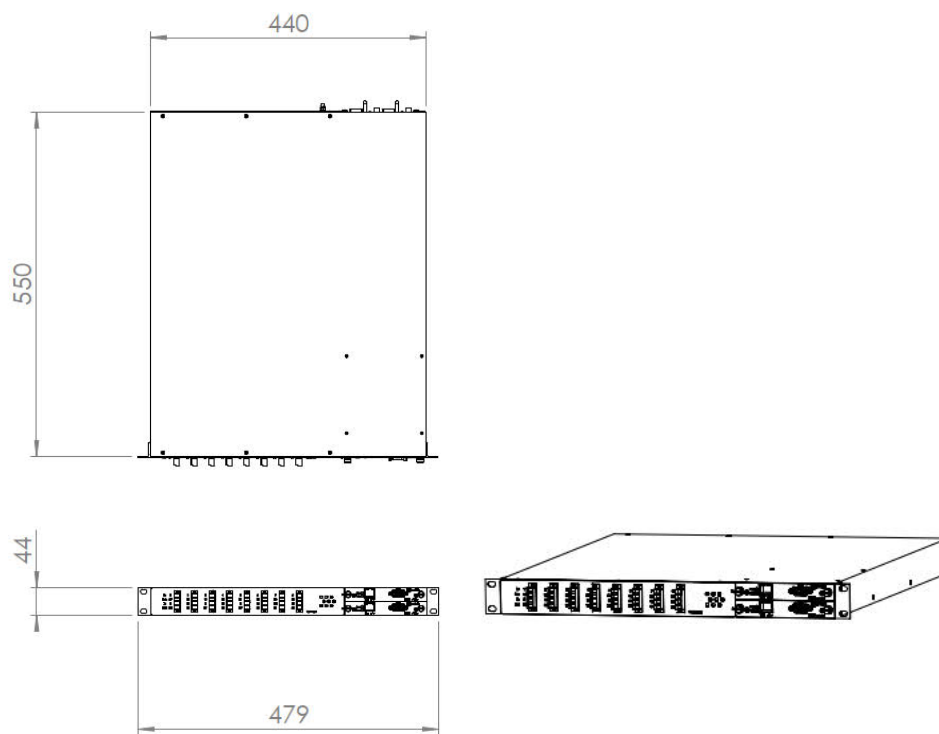


Figure 1 Dimensions of Series 6000 1RU Multimode switch

The Polatis Series 6000 Multimode 2RU switch configuration has dimensions in millimeters: 550 deep x 440 wide x 88mm (2RU) tall, with front mounting ears 479 mm in width. (21.7 x 17.3 x 3.5 inches with front mounting ears 19" in width). The switch weighs 12kg (26.5 pounds). **The switch must be installed on a solid surface capable of supporting 15kg (33 pounds) and cannot be suspended from only the front panel mounting ears.**

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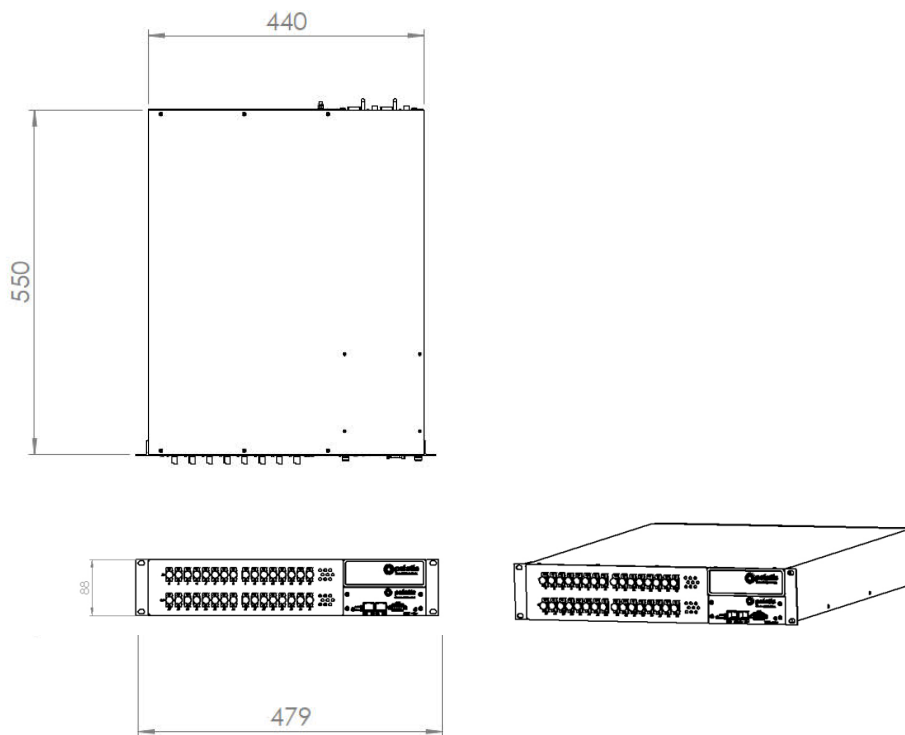
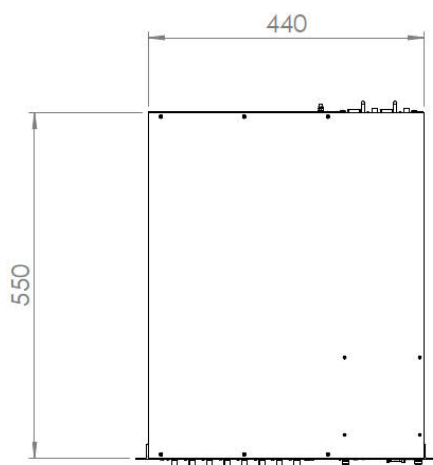


Figure 2 Dimensions of Series 6000 2RU Multimode switch

The Polatis Series 6000 Multimode 3RU switch configuration has dimensions in millimeters: 550 deep x 440 wide x 132mm (3RU) tall, with front mounting ears 479 mm in width. (21.7 x 17.3 x 5.25 inches with front mounting ears 19" in width). The switch weighs 13kg (28.7 pounds). **The switch must be installed on a solid surface capable of supporting 15kg (33 pounds) and cannot be suspended from only the front panel mounting ears.**



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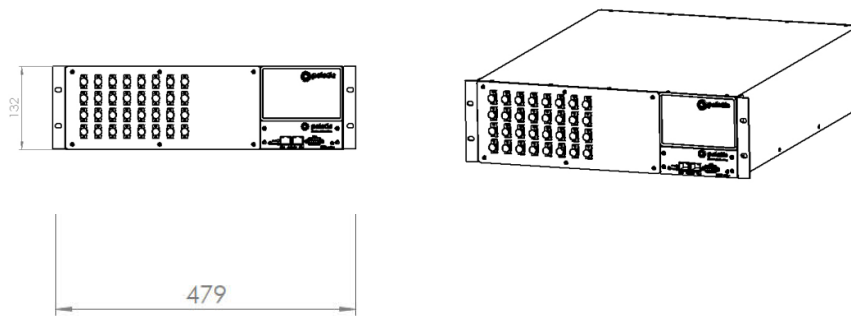


Figure 3 Dimensions of Series 6000 3RU Multimode switch

The Series 6000 Multimode switches are shipped in a shock absorbing dual cardboard enclosure with foam shock isolators.

Please retain all packaging, as any shipment of the switch must use this original packaging to ensure the switch is not damaged in transit. The overall dimensions of the outer box are 89 x 69 x 38cm (35 x 27 x 15") and the packaging weighs 7.7 kg (17 pounds).



6 Mounting and installation

6.1 Mounting

The POLATIS switch is designed to be installed in a standard 19" equipment rack and must be supported by a shelf from below. It is not designed to be supported by the front mounting ears only. Failure to properly install the switch on anything but a secure shelf or platform will void the warranty.

6.2 Unpacking and installing

This piezo-electric switch is robustly designed but can be damaged if it is dropped or subjected to excessive shock. Please take care in handling the switch.

All packaging materials including the outer box, foam shock absorbers, inner box, foam inner-box lining and plastic or foil bag enclosing the switch must be saved for any future shipment.

1. Open the outer box.
2. Remove and save the 4 foam corner shock isolators between the top of the inner box and the outer box.
3. Open the inner box and carefully open the foil liner bag by lifting the edge of the top sheet and detaching the adhesive from the bag underneath.
4. Remove the foam top sheet(s).
5. Place the switch on a solid surface capable of supporting 15kg (33 pounds).

The switch must be properly installed on a solid surface, rack, or rails capable of supporting 15 kg (33 pounds).

6. Once this is done, the front mounting ears may be secured to the rack to prevent switch movement, but as stated before, **the switch weight must be supported from the bottom by a secure platform.**

6.3 Ground (earth) connections

The switch has a ground (earth) lug located below the green earth symbol. Please connect this to a common ground (earth) connection that is shared by the AC power feeds.



Please use the appropriate ring or spade connector to a ground conductor of at least 18 AWG wire, following industry-standard grounding practices. The table below specifies the recommended wire and crimp terminal.

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For permanently connected equipment provided with terminal(s) for connection to mains supply, reference is made to the national building wiring requirements for the size of the protection earthing (grounding) conductor.

Table 3 Ground (Earth) connection recommended wire and terminal specification

Description	Type	Approval
Wire	18 AWG Green or Green+Yellow	NRTL
Terminal	Spade 16-22 AWG	NRTL

6.4 Power connections

The switch has two redundant hot-swap Power Supply Units (PSUs). It may be powered by just one of these PSUs and will operate in a redundant fashion when both are powered. Should an issue occur with one of the supplies, the other will continue to operate the unit without interruption while the unit with an issue can be replaced.

It is suggested that if a redundant power distribution scheme is used, that the primary power connection be made to the left PSU, labeled PSU 1, and the backup power connection be made to the right PSU, labeled PSU 2.

PSU 1 PSU 2



Power connection for the AC module is via an integral IEC socket. Power connection for the DC module is via screw terminals, as shown in the sections below:

6.4.1 Type AC (100-240VAC) PSUs

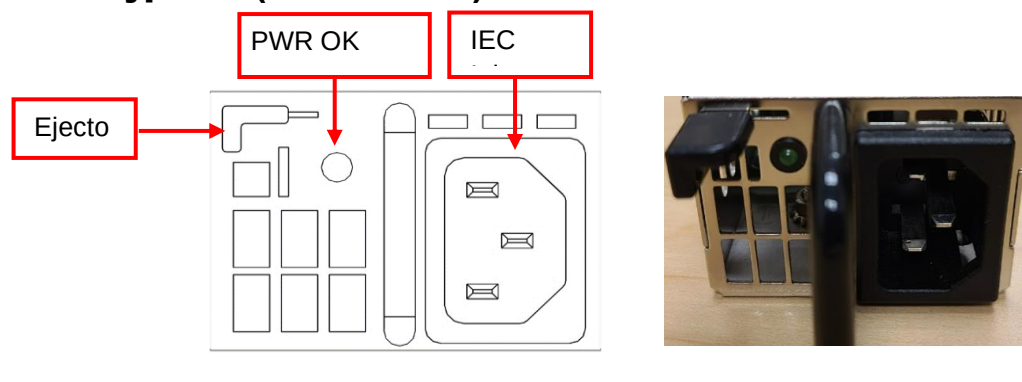
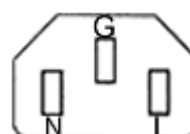


Figure 4 Type AC (100-240VAC) PSUs

For the 100-240VAC Autosensing PSUs with C14 power connector above, the mating IEC320 C13 connector shown below is used with the following connections:

- G – Ground (Earth)
- L – Line 100-240VAC (47-63 Hz)
- N – Neutral



IEC320 C13

6.4.2 DC PSU Nominally -48VDC (-40 to -72VDC) PSUs

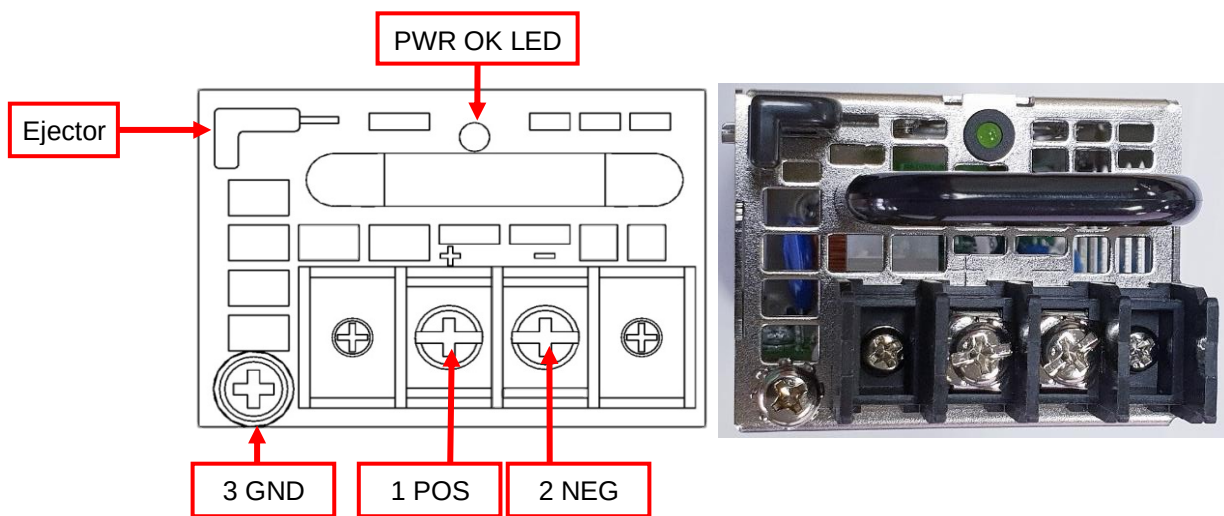


Figure 5 6.4.2 DC PSU Nominally -48VDC (-40 to -72VDC) PSUs

For the Type T Nominally -48VDC PSUs, the following connections are made:

Table 4 Type T Nominally -48VDC PSU connections

Description	Screw Terminal Pin
Ground (Earth)	3
-48VDC	2
Return	1

Appropriately rated cable and terminals should be used for the installation. We recommend using the following wire and crimp terminal. The cabling is not supplied with the product.

Table 5 Recommended wire and crimp terminal for installation

Description	Type	Approval
Wire	18 AWG	NRTL
Terminal	Spade 16-22 AWG	NRTL

6.5 Spares

Spare PSU modules can be ordered from HUBER+SUHNER Polatis customer support. The part codes are shown in Table 9 below:

Table 6 PSU part codes

PSU Module	Zippy Part Code	Polatis Part Code
AC	G1X-1060V12	85135215 (P-1002-2382)
DC	DG1X-1060V12	85114741 (P-1002-2053)

6.6 Operation

Normal operation assumes that both hot-swap redundant power supplies are installed and receiving external power, whether AC or DC. However, the optical switch will operate properly with just one supply installed and powered.

When only one of the two PSU modules is installed and powered by external AC or DC mains, or there is a failure of one of the PSUs, the switch will emit a warning alarm tone indicating that one PSU is not installed, not powered, or has failed.

This alarm may be reset and silenced by using a thin probe, such as a pen, to press the recessed red button just to the right of the hot-swap power supply cage, adjacent to the chassis power switch. This is shown below.



PSU Audible Alarm Reset Button

The alarm will remain silenced for as long as power is maintained on the single PSU. Once a second PSU is inserted and powered, the alarm will remain off, but will again sound if either of the two PSUs are not operational or loose external power.

If the alarm is not silenced manually by pressing the red button, once both PSUs are properly installed and powered, the alarm will automatically silence.

When removing and inserting hot-swap power modules, please observe the following precautions:

1. You should only remove a power supply module after you have disconnected the mains lead from the AC IEC inlet, or the screw terminals on the DC PSU inlet.
2. Only reconnect the mains lead after fully inserting the power supply unit module.
3. Do not remove and immediately re-insert a power supply module. Allow at least 5 seconds for any residual power to discharge before fully inserting the power supply unit module.

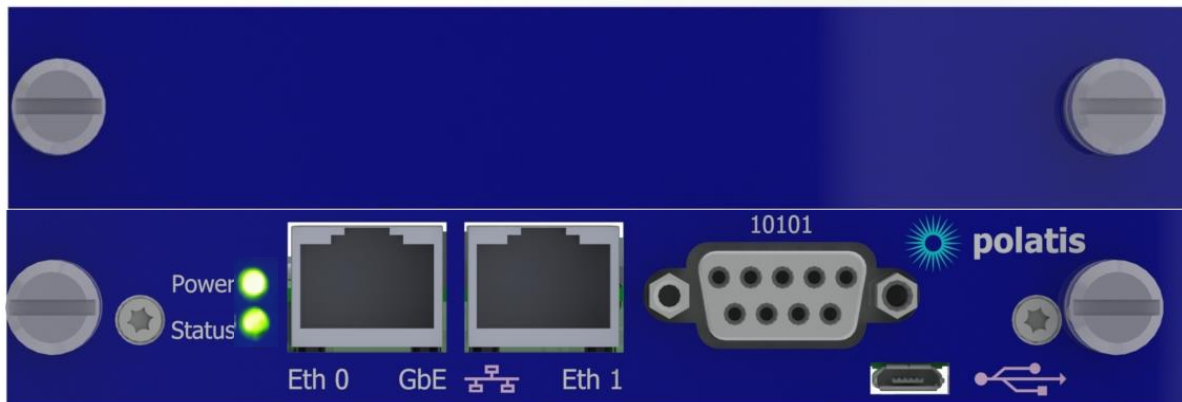
6.7 Control interfaces

The switch has serial and Ethernet control interfaces available on the blue Network Interface Controller(s) (NIC(s)).

The Series 6000 Multimode switch may be ordered with the “D” option for dual multisession Ethernet & serial or the “H” option for two hot-swap NICs. The NIC is located at the right of the front panel. The standard configuration for the D option on a switch like S-OST-16x16-

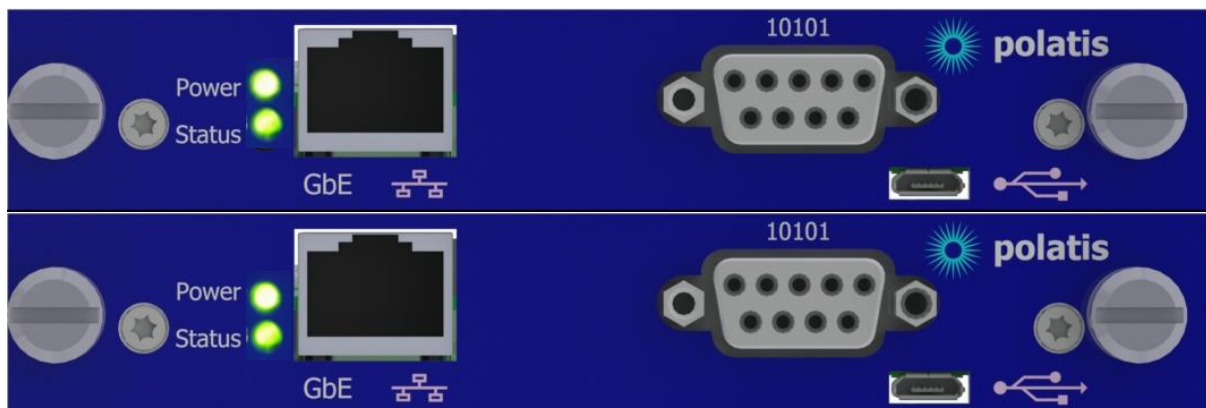
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LP9-DMHNS will have a single NIC with two RJ-45 Ethernet connections and a DB-9 RS-232 Serial port as shown below:



You may connect to the DB-9 RS-232 Serial port connector, and the two RJ-45 Ethernet connectors. The default IPv4 address for the left RJ-45 Ethernet port 0 is 192.168.0.1. The default IPv4 address for the right RJ-45 Ethernet port 1 is 192.168.1.1. In this illustration, the POLATIS switch Network Interface Controller (NIC) is shown with the power and status LEDs, RJ-45 Ethernet port 0 and 1 connectors and serial RS-232, DB-9 connector. Also shown is a micro-USB connector for diagnostic use.

The standard configuration for the H option on a switch like S-OST-16x16-LP9-HMHNS will have two NICs with a single RJ-45 Ethernet connection and a DB-9 RS-232 serial port on each as shown below:



You may connect to the DB-9 RS-232 serial port connector on the lower NIC in slot A, and both RJ-45 Ethernet connectors. The default IPv4 address for the NIC in the lower position A is Ethernet port 0 is 192.168.0.1.

The default IPv4 address for the NIC in the upper position Ethernet port 1 is 192.168.1.1. In this illustration, the POLATIS switch Network Interface Controllers (NICs) are shown with the power and status LEDs, RJ-45 Ethernet connector, serial RS-232, DB-9 connector, and a diagnostic micro-USB connector.

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With the hot-swap NICs, one NIC takes the role of the active NIC, and the status LED shows solid green. The second NIC takes on the role of the standby NIC, and the status LED flashes green to indicate that this is working in standby mode. When initially powered up, NIC A in the bottom slot, takes the role of the active NIC, and NIC B in the top slot takes the role of the standby NIC. If NIC A is subsequently removed or rebooted, NIC B becomes the active NIC after about 10 seconds, and NIC A will assume the role of the standby NIC when reinserted or rebooted. The status LEDs will change accordingly.

6.7.1 Serial RS-232 connector

The DB-9 RS-232 serial connector is configured as follows. The pinout of the male DB-9 connector which mates with the switch is shown:

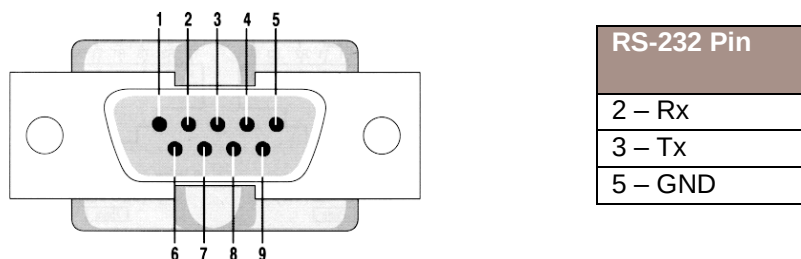


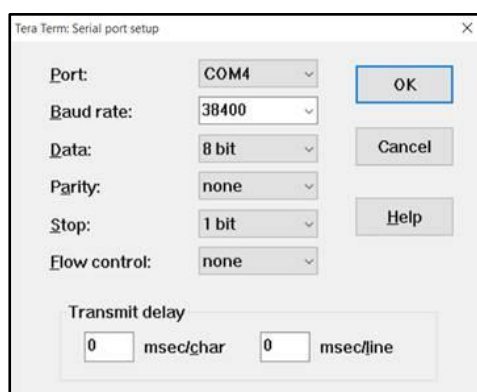
Figure 6 Serial RS-232 connector

Note: RS232 3-wire interface is sufficient. (CTS/RTS are not used.)

The RS-232 serial port settings are as follows.

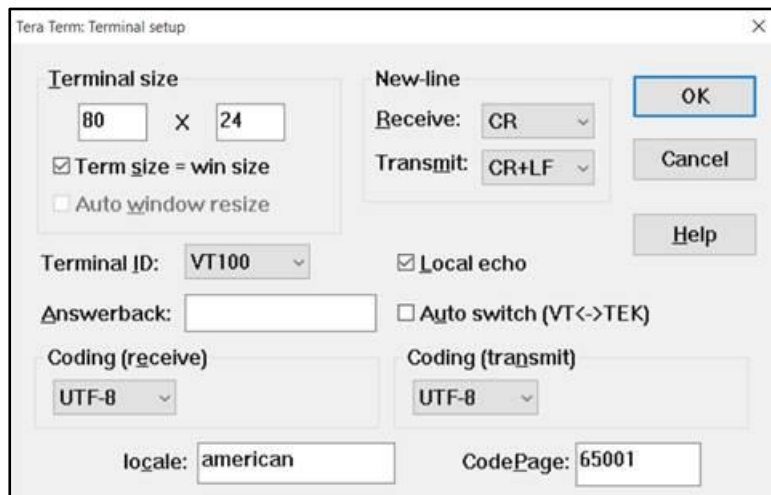
Note: The example below shows COM4, but please set the COM port correctly for your PC (Often COM1 for a DB-9 serial port) or the correct COM port for a USB to serial adapter.

These examples use the Tera Term Pro Terminal Emulator program, but you may use other programs such as Putty or HyperTerminal.



There is no SCPI prompt, and please verify **Local echo** is turned on to be able to see the SCPI commands entered. This is shown in the Tera Term Pro example below where **Local echo** is checked and the “CR+LF” New-line setting is specified.

HUBER+SUHNER POLATIS® Series 6000 Multimode Optical Switch User Manual



6.7.2 Ethernet interface

The switch is configured with two multi-session Ethernet interfaces, and each can support up to 20 concurrent sessions. The Web GUI is available by directing your web browser to the IP address of the switch.

- For the left or lower RJ-45, Ethernet port 0, the IPv4 address defaults to 192.168.0.1.
- For the right or upper RJ-45, Ethernet port 1, the IPv4 address defaults to 192.168.1.1.

If you set your PC to an IPv4 address on the same subnet as these defaults, 192.168.0.10 for example, you can connect to the POLATIS switch Ethernet port 0 with a simple Cat5 cable without any hub, switch or cross-over cable.

1. Please verify you can ping 192.168.0.1 from the PC.
2. Then direct your web browser to <https://192.168.0.1>.
3. Ignore any security certificate warnings, and you should be presented with the POLATIS Web GUI.

This will allow you to control all aspects of each switch and change all IP settings to those desired to connect to your network. Details are covered in [Section 7 Operating the switch](#).

- The SCPI Command Line Interface, which is available on the serial port, is also available on Ethernet port 5025.
- The TL1 Command Line Interface is available on Ethernet port 3082.
- SNMP, NETCONF and RESTCONF are available via the Ethernet interface.

Please see [Section 2](#) for the details on the user manuals which provide complete coverage of these command interfaces.

6.8 Optical connections

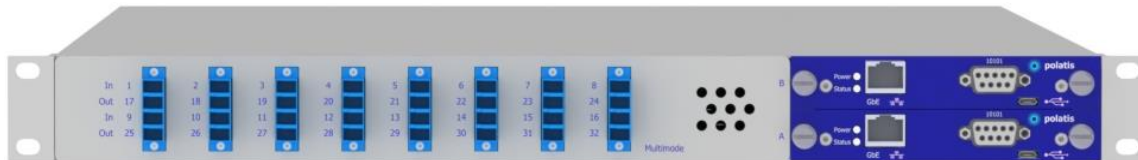
The Polatis 8x8 Multimode switch uses the following naming convention: Input ports are labelled 1 through 8 and output ports are labelled 9 through 16.

The Polatis 16x16 Multimode switch uses the following naming convention: Input ports are labelled 1 through 16 and output ports are labelled 17 through 32.

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For the 16x16 Multimode switch using LC/UPC Multimode optical connectors, the port configuration is shown below.

The first input port 1 is matched with the first output port 17, and in a similar fashion, all other input ports are matched with a corresponding output port, up through the last input port 16 matched with the last output port 32.



The LC/PC Multimode ports on the switch shown above are configured to support individual LC/PC Multimode connectors, or duplex LC/PC Multimode fiber connector with Tx/Rx pairs as shown below.



7 Operating the switch

7.1 Power-up sequence

Once the two power connections are made, and the Ethernet and/or serial interfaces are connected, please power up the switch by turning on the two power switches under PSU 1 and PSU 2 on the back panel.

7.2 Power and status LEDs

Each NIC has two LEDs indicating power and status. Whenever one of the PSUs is turned on, the power LEDs will illuminate green.

Immediately after the power is turned on, the switch begins an extensive Power-On-Self-Test (POST) sequence. Depending on the switch, this POST sequence may take a full 7 minutes. During this time, the status LEDs will transition from off to amber, then to off and finally will illuminate a solid green once the NIC completes booting. The switch still needs up to 6 minutes to complete its POST sequence.

With the hot-swap NICs, one NIC takes the role of the active NIC, and the status LED shows solid green. The second NIC takes on the role of the standby NIC, and the status LED flashes green to indicate that it is working as the standby NIC.

When initially powered up, NIC A in the bottom slot, takes the role of the active NIC, and NIC B in the top slot takes the role of the standby NIC. If NIC A is subsequently removed or rebooted, NIC B becomes the active NIC after about 10 seconds, and NIC A will assume the role of the standby NIC when reinserted or rebooted. The status LEDs will change accordingly.

7.3 Web Graphical User Interface (GUI)

Once the status LED is green, the Web GUI is available by directing your web browser to the IP Address of the switch.

- For the primary Ethernet port 0, this defaults to 192.168.0.1.
- For the right or upper Ethernet port 1, this defaults to 192.168.1.1.

If you set your PC to something on the same subnet as these defaults, 192.168.0.10 for example, you can connect to the POLATIS switch Ethernet port 0 with a simple Cat5 cable without any hub, switch or cross-over cable.

1. Please verify you can ping 192.168.0.1 from the PC.
2. Then direct your Web Browser to <https://192.168.0.1>.
3. Please ignore any security certificate warnings, and you should be presented with the Polatis Web GUI.

This will allow you to control all aspects of each switch and change all IP settings to those desired to connect to your network.

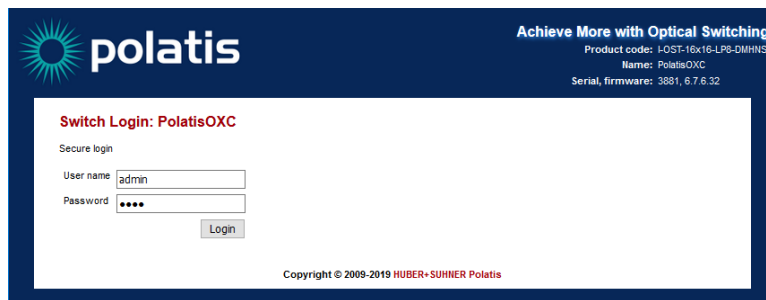
For a dual-Ethernet card, the primary Ethernet port 0 is the left socket, and the secondary Ethernet port 1 is the right socket. For the hot-swap NIC, the Ethernet address stays with the card slot or position, rather than the NIC.

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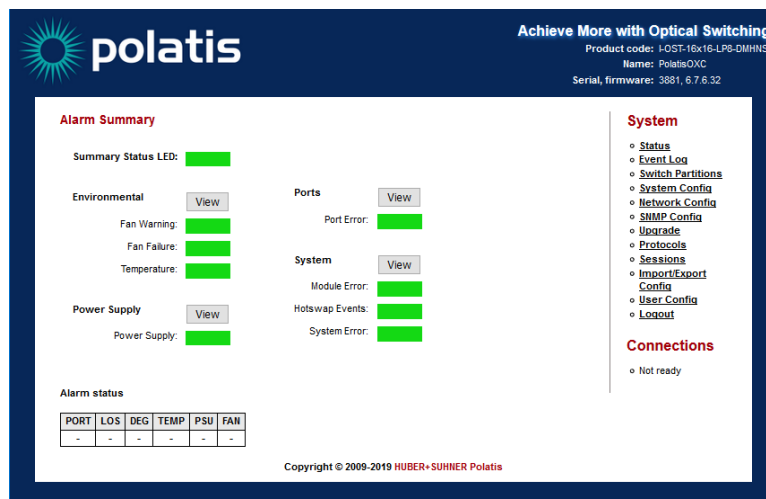
The primary Ethernet Port 0 address is available on the NIC in the lower position, and the secondary Ethernet Port 1 address is available on the card in the upper position.

In the following examples, the Web GUI is shown for a 16x16 Series 6000 Multimode switch.

1. The default login to the Web GUI is **admin (or root)**, and the default password is also **root** (both all lowercase).

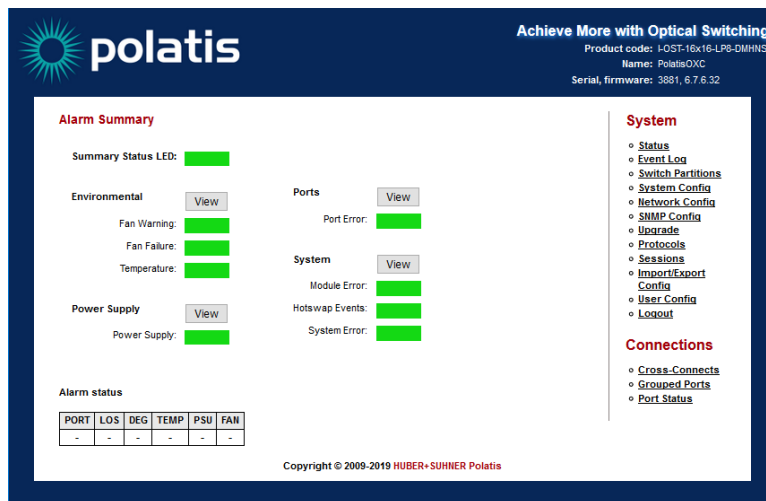


2. Select **Login**, which will take you to the status screen.



3. All the Web GUI functional links are shown on the right side of the screen. Under **Connections**, the entry "**Not ready**" indicates the switch is still completing its POST sequence. Please wait until the POST sequence completes and the following screen is shown:

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The following links are shown on the right:

7.3.1 System

- Status
- Event Log
- Switch Partitions
- System Config
- Network Config
- SNMP Config
- Upgrade
- Protocols
- Sessions
- Import/Export Config
- User Config
- Logout

7.3.2 Connections

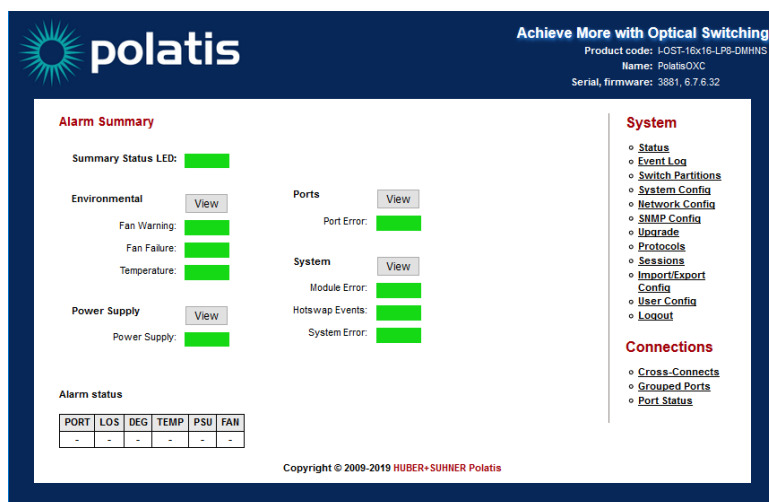
- Cross-Connects
- Grouped Ports
- Port Status

7.3.3 Status screen

The **Status** screen shown above provides a status snapshot of the switch. Normally all bars be shown in green, and any warnings will be shown in amber, and any failures in red. The only expected variation from the screen shown with all green bars is when only one of the two redundant hot-swap Power Supply Units (PSUs) is turned on. In this case, the power supply bar will appear amber.

To see the details on any events shown on the Status Screen, Select the **View** button next to that section, or select the **Events Log** link on the right.

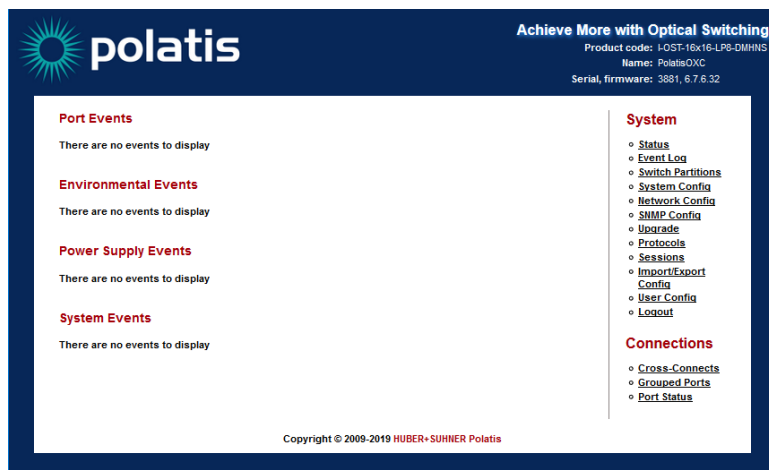
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7.3.4 Event Log screen

The **Event Log** screen shown below provides a detailed summary of any switch events which may affect the green status indicators on the **Status** screen. Events are listed under the four categories of:

- Port Events
- Environmental Events
- Power Supply Events
- System Events



Under normal operating conditions, there should not be any events listed, except under Power Supply Events, where an amber warning indicates that one of the hot-swap redundant power supplies is turned off, not plugged in, or not operational. In this case you may see the following power supply events listed on the event log. The NIC status LED will turn amber.

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The screenshot shows the Polatis web interface with the following sections:

- Port Events:** "There are no events to display"
- Environmental Events:** "There are no events to display"
- Power Supply Events:** A table with one event:

ID	Sub Switch	Timestamp	Event Type	Message	Select
1	N/A	2021 Mar 28 at 16:50:29.024	PSU	The PSU (psu1) has been switched off.	☐

Buttons below the table: "Select All Events", "Delete Selected Events...", "Clear Selected Events".
- System Events:** "There are no events to display"
- System Menu:** Status, Event Log, Switch Partitions, System Config, Network Config, SNMP Config, Upgrade, Protocols, Sessions, Import/Export Config, User Config, Logout.
- Connections Menu:** Cross-Connects, Grouped Ports, Port Status.

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Events are initially shown in amber, or red and may be selected and cleared or deleted.

This screenshot is identical to the previous one, but the "Clear Selected Events" button in the "Power Supply Events" section is highlighted with a blue border, indicating it is the next step in the process.

After selecting the checkbox after one or more events, choose **Clear Selected Events** to clear the amber color but retain the event in the log.

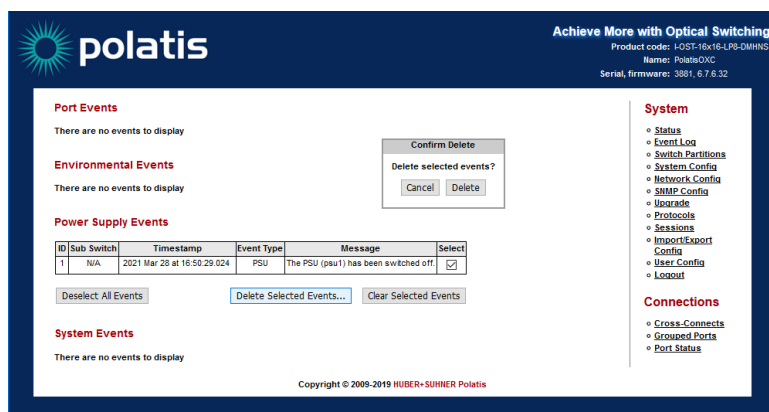
The screenshot shows the Polatis web interface after the "Clear Selected Events" action. The "Power Supply Events" table now shows the event with a grey background and the "Select" checkbox is unchecked.

ID	Sub Switch	Timestamp	Event Type	Message	Select
1	N/A	2021 Mar 28 at 16:50:29.024	PSU	The PSU (psu1) has been switched off.	☐

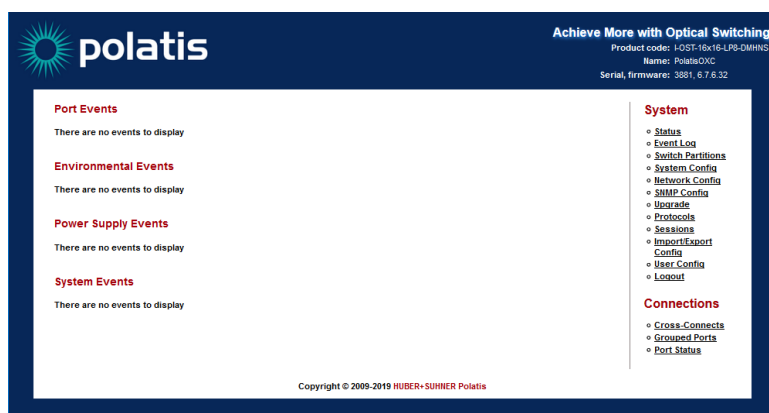
Buttons below the table: "Select All Events", "Delete Selected Events...", "Clear Selected Events".

To delete the event from the log, select it and choose **Delete Selected Events**, and confirm the deletion in the pop-up.

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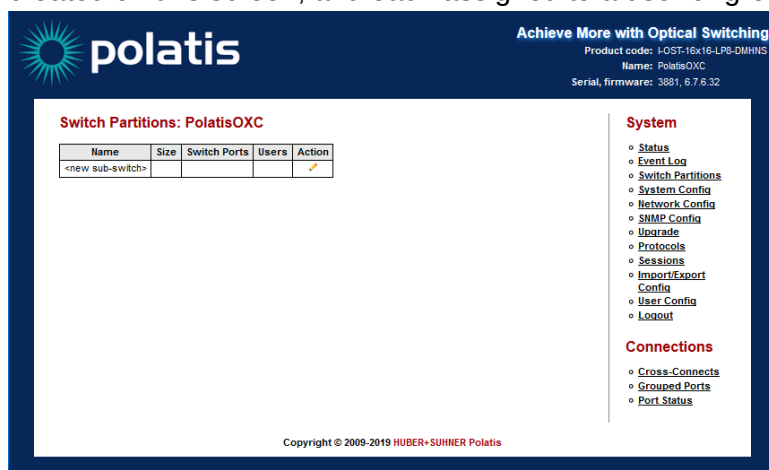


Now all selected events are deleted from the log.



7.3.5 Switch Partitions screen

The **Switch Partitions** Screen allows sub-switch partitions to be created. First, users have to be created using the **User Admin** screen, and then new sub switch partitions can be created on this screen, and each assigned to a user or group of users.



7.3.6 System Config screen

The **System Config** screen allows configuring basic attributes of the system including:

- **Switch Name:** an assigned name reported on Web GUI screens.
- **Switch Startup Mode:**
 - **Dark:** no connections made when powered up.

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- **Last Known State:** If the switch is power cycled, once it completes the boot and Power-On-Self-Test (POST) sequence, it resumes all connections that were made when last powered.
- **Session Timeout Period:** Number of minutes which Administrative or User Web GUI sessions will remain open after the last keyboard interaction. To remain open indefinitely, set this value to zero.
- **Web Page Refresh time:** number of seconds between Web GUI screen refreshes.
- **System Date and Time:** Sets the system date and time. This needs to be reset after switch is powered down for more than 1 minute. NTP Servers may be specified to automatically synchronize the switch settings.
- **Remote Syslog:** Machine IP Address and log file name for remote system logging.
- **Reset Network Card:** Resets the Network Interface Controller (NIC) card without affecting existing optical connections through switch.

The screenshot shows the 'System Configuration: PolatisOXC' web interface. The header includes the Polatis logo and the slogan 'Achieve More with Optical Switching'. It also displays product code (LOST-10x16-LP8-OMHNS), name (PolatisOXC), and serial/firmware (2881, 6.7.6.32). The main configuration area is divided into sections: Switch Name (PolatisOXC), Switch Startup Mode (OXC Mode: Dark), Session Timeout Period (User Sessions: 30, Admin Sessions: 15), Web Page Refresh Period (10), System Date and Time (Uptime: 00:15:02, Date: 27/03/2021, Time: 20:07), NTP servers (table with columns: Address, Min Poll, Max Poll, NTP version, Action), Activity Log (checkboxes for Cross-Connects, Optical Power Monitors, System parameters, Authentication and Authorisation), Remote syslog (Address, Facility: LOG_USER), Hardware reports (Chassis fans table with Fan Speed (%)), and Network Card Reboot (Reboot Network Card button). A sidebar on the right lists navigation links under 'System' (Status, Event Log, Switch Partitions, System Config, Network Config, SNMP Config, Upgrade, Protocols, Sessions, Import/Export Config, User Config, Logout) and 'Connections' (Cross-Connects, Grouped Ports, Port Status). The footer shows 'Copyright © 2009-2019 HUBER+SUHNER Polatis'.

7.3.7 Network Config screen

The **Network Config** screen allows configuring the internet protocol settings for the system including:

- IP Address
- Gateway
- Subnet Mask
- Broadcast Address

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Any changes to these values will not take effect until the Network Interface Controller (NIC) is reset, or the switch is power cycled. When the switch is power cycled, optical connections are lost.

To reset the NIC card without power cycling the switch, to insure existing optical connections are not affected, select the checkbox above the **Submit** button before clicking the submit button. This will only reset the NIC and not affect existing optical traffic. The Web GUI will not be responsive and existing optical connections cannot be modified while the NIC is rebooting.

Network Configuration: PolatisOXC

	Ethernet 0	Ethernet 1
MAC Address	00:bd:2e:01:08:85	00:bd:2e:01:08:86
IP Address	192.168.0.1	192.168.1.1
Gateway	0.0.0.0	0.0.0.0
Use Default Subnet	☐	☐
Subnet	255.255.255.0	255.255.255.0
Use Default Broadcast Address	☐	☐
Broadcast Address	192.168.0.255	192.168.1.255

☐ Check this box to restart the network interface after configuring the network settings

Submit

Note: The network settings are not activated until the network card is reset.

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System

- Status
- Event Log
- Switch Partitions
- System Config
- Network Config
- SNMP Config
- Upgrade
- Protocols
- Sessions
- Import/Export Config
- User Config
- Logout

Connections

- Cross-Connects
- Grouped Ports
- Port Status

7.3.8 SNMP Config screen

The **SNMP Config** Screen allows configuring SNMP Traps.

SNMP Configuration: PolatisOXC

Download SNMP MIBs [here](#).

Notifications

When this tab is used to configure notifications, all events in the switch will send notifications to the addresses entered below.

Notification Type:

Destination	Community/V3 user	SNMP Version	V3-inform data	
<new row>				
<new row>				
<new row>				
<new row>				
<new row>				

Configure SNMP

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System

- Status
- Event Log
- Switch Partitions
- System Config
- Network Config
- SNMP Config
- Upgrade
- Protocols
- Sessions
- Import/Export Config
- User Config
- Logout

Connections

- Cross-Connects
- Grouped Ports
- Port Status

1. Select the yellow pencil to enter information regarding the SNMP trap.

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SNMP Configuration: PolatisOXC

Download SNMP MIBs [here](#).

Notifications

Simple Configuration | Full Configuration

When this tab is used to configure notifications, all events in the switch will send notifications to the addresses entered below.

Notification Type:

Destination | **Community/V3 user** | **SNMP Version** | **V3-inform data**

Destination	Community/V3 user	SNMP Version	V3-inform data
192.168.0.10	private	v3 Auth: yes, Priv: yes	☒
<new row>			☐
<new row>			☐
<new row>			☐
<new row>			☐

System

- Status
- Event Log
- Switch Partitions
- System Config
- Network Config
- SNMP Config
- Upgrade
- Protocols
- Sessions
- Import/Export Config
- User Config
- Logout

Connections

- Cross-Connects
- Grouped Ports
- Port Status

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- After entering the trap information click **OK**, and the trap will be entered.

SNMP Configuration: PolatisOXC

Download SNMP MIBs [here](#).

Notifications

Simple Configuration | Full Configuration

When this tab is used to configure notifications, all events in the switch will send notifications to the addresses entered below.

Notification Type:

Destination | **Community/V3 user** | **SNMP Version** | **V3-inform data**

Destination	Community/V3 user	SNMP Version	V3-inform data
192.168.0.10	private	v3 Auth: yes, Priv: yes	☒
<new row>			☐
<new row>			☐
<new row>			☐
<new row>			☐

System

- Status
- Event Log
- Switch Partitions
- System Config
- Network Config
- SNMP Config
- Upgrade
- Protocols
- Sessions
- Import/Export Config
- User Config
- Logout

Connections

- Cross-Connects
- Grouped Ports
- Port Status

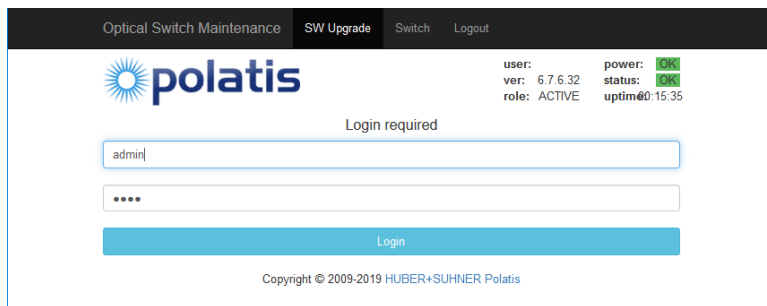
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- Click **Configure SNMP** to activate the changes.

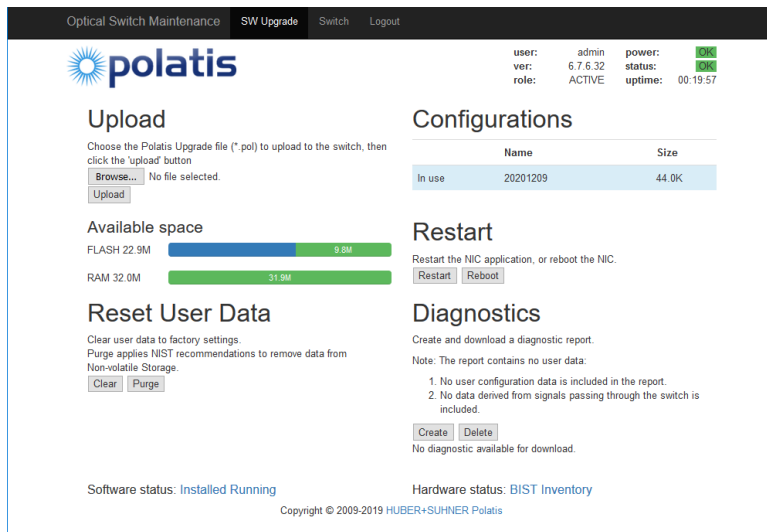
7.3.9 Upgrade screen

The Upgrade screen provides for upgrading the POLATIS switch software, creating diagnostic logs and forcing a restart or reboot of the NIC cards without affecting optical cross connections.

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This screen uses the administrative login **admin** with the password **root**.



Under the **Upload** heading, the Upgrade screen allows browsing for and uploading POLATIS software updates or serial number specific NIC configuration files.

All POLATIS software upgrade packages will be provided with full instructions for performing software and configuration changes.

Under the **Configurations** heading, it allows selecting the serial number specific NIC configuration file.

- The **Reset User Data** allows for restoring the switch to the factory delivered configuration, removing all customer entered settings.
- The **Restart** button causes the NIC application to restart, and the **Reboot** button completely reboots the NIC, including the underlying Linux Operating System.
- The **Diagnostics** Section allows for creating and downloading diagnostic files for helping to remotely troubleshoot any switch problems. These can be forwarded to HUBER+SUHNER Polatis for analysis.

 **Note:** Do not login to the Upgrade screen and load new software or make changes to the configuration without contacting HUBER+SUHNER Polatis Customer Support. Unauthorized changes made through the upgrade screen may render the switch inoperative and it may have to return to HUBER+SUHNER Polatis for repair.

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7.3.10 Protocols screen

Switches with a minimum 6.X software release feature a **Protocols** link under the **System** heading. This allows administrative users (such as the default login **admin** (or **root**) with password **root**) to configure the active protocols supported by the switch.

polatis Achieve More with Optical Switching
Product code: LOST-16x16-LP8-DMHNS
Name: PolatisOXC
Serial, firmware: 3881, 6.7.6.32

Protocols Configuration

Check or uncheck the protocols to enable or disable them, then click the 'Apply' button for the changes to take effect

Admin

- ☒ SCPI (console,38400,8N1)
- ☒ SCPI (tcp,333)
- ☒ SCPI (tcp,5025)
- ☐ TL1 (console,38400,8N1)
- ☒ TL1 (tcp,3082)
- ☒ TL1 (tcp,6252)
- ☒ TL1 (usbserial,38400,8N1)
- ☐ SNMPV1 (udp,161)
- ☐ SNMPV2C (udp,161)
- ☒ SNMPV3 (udp,161)
- ☒ NETCONF (tcp,830)
- ☒ ENG (tcp,9999)
- ☒ RESTCONF (tcp,8008)

System

- [Status](#)
- [Event Log](#)
- [Switch Partitions](#)
- [System Config](#)
- [Network Config](#)
- [SNMP Config](#)
- [Upgrade](#)
- [Protocols](#)
- [Sessions](#)
- [Import/Export Config](#)
- [User Config](#)
- [Logout](#)

Connections

- [Cross-Connects](#)
- [Grouped Ports](#)
- [Port Status](#)

Note

Protocol settings are not changed immediately.

- SSH protocols (TL1/6252 and NETCONF/830) require a reboot to change.
- HTTP protocols (HTTP/80 and HTTPS/443) require a reboot to change (but can only be configured from a different protocol interface).
- Other changes require the application to be restarted.

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This lists all protocols that are supported by the switch, each adjacent to a check box. Normally, these are all checked, and the interfaces are all enabled. If an administrative user wishes to deactivate an interface, the checkbox should be unchecked and then select the **Apply** button. The change will only take effect after the NIC is restarted from the System Config or Network Config screens.

Below is a list of available protocols:

Table 7 Available protocols

Protocol	Description
SCPI (console,38400,8N1)	SCPI on RS-232 serial port DB-9 connector (38400 baud, 8-bits, no parity, 1 stop bit, no flow control, Local Echo On, New Line TX=CR+LF)
SCPI (tcp,333)	Diagnostic SCPI on Ethernet port 333 (for POLATIS diagnostics)
SCPI (tcp,5025)	SCPI on Ethernet port 5025
TL1 (usbserial,38400,8N1)	Diagnostic TL1 on USB or micro-USB (for POLATIS diagnostics)
TL1 (tcp,3082)	TL1 on Ethernet port 3082 (secure, requires logon)
SNMPV1 (udp,161)	SNMP V1 using UDP on port 161 (all checked SNMP protocols are available simultaneously)
SNMPV2C (udp,161)	SNMP V2C using UDP on port 161 (all checked SNMP protocols are available simultaneously)
SNMPV3 (udp,161)	SNMP V3 using UDP on port 161 (all checked SNMP protocols are available simultaneously)
NETCONF (tcp,830)	NETCONF on Ethernet port 830
ENG (tcp,9999)	Diagnostic engineering interface for Polatis use only on Ethernet port 9999
RESTCONF (tcp,8008)	NETCONF on Ethernet port 8008

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Note: The POLATIS boot sequence is normally shown on the RS-232, DB-9 console serial port. This normally ends with the SCPI READY message as shown on the screen below. The *idn? identification command was then typed in and executed in this example.

```
COM4 - Tera Term VT
File Edit Setup Control Window Help
Starting g_serial module (USB)
Starting softdog module
enable LEDs
Stopping Bootlog daemon: bootlogd.

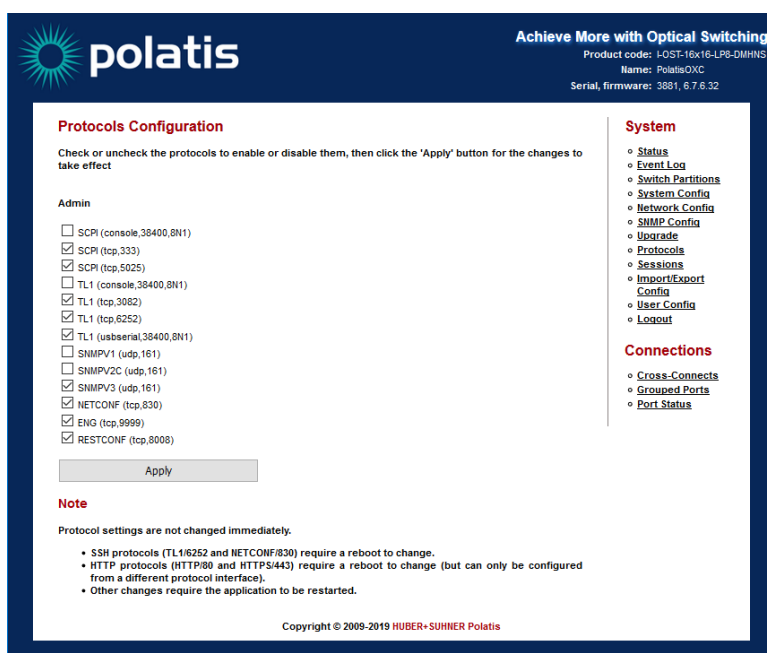
POLATIS

S/N 3881
Ver 6.7.6.32
eth0 inet 192.168.0.1/24 brd 192.168.0.255 scope global eth0
eth1 inet 192.168.1.1/24 brd 192.168.1.255 scope global eth1

Sat Mar 27 20:15:14 UTC 2021
Configuring...
SCPI READY

*idn?
Polatis,I-OST-16x16-LP8-DMHNS,003881,6.7.6.32
```

If the first SCPI entry is unchecked, and the apply button selected as shown below, and the switch is restarted, you will see the boot sequence, but never get the SCPI READY message and the serial port will not accept nor respond to SCPI commands.



Boot sequence ending without SCPI enabled on serial port.

```
COM4 - Tera Term VT
File Edit Setup Control Window Help
Starting backlight module
Starting g_serial module (USB)
Starting softdog module
enable LEDs
Stopping Bootlog daemon: bootlogd.

POLATIS

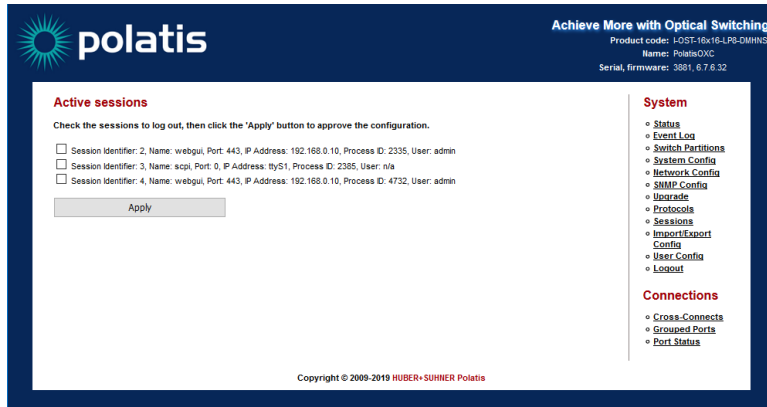
S/N 3881
Ver 6.7.6.32
eth0 inet 192.168.0.1/24 brd 192.168.0.255 scope global eth0
eth1 inet 192.168.1.1/24 brd 192.168.1.255 scope global eth1

Sat Mar 27 19:52:56 UTC 2021
Configuring...
```

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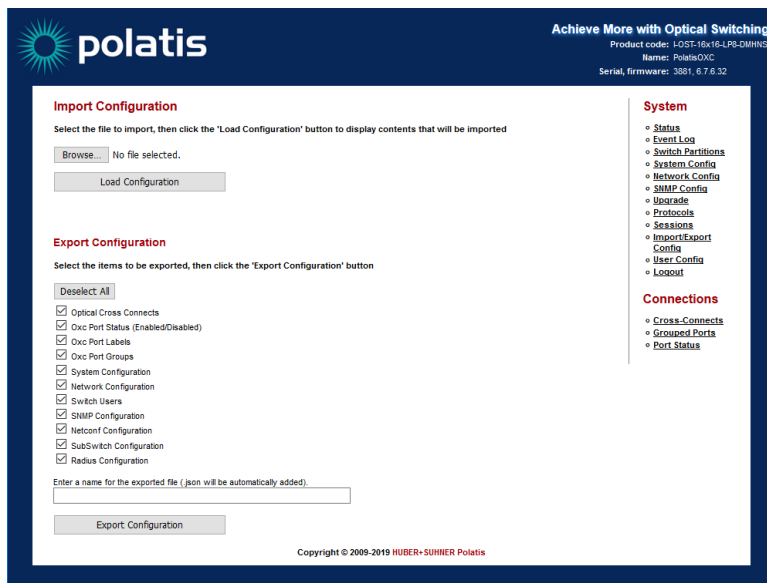
7.3.11 Sessions screen

The sessions screen shows the active user sessions. Checking the check box to the left of a session and clicking **Apply** will delete that user session.



7.3.12 Import/Export screen

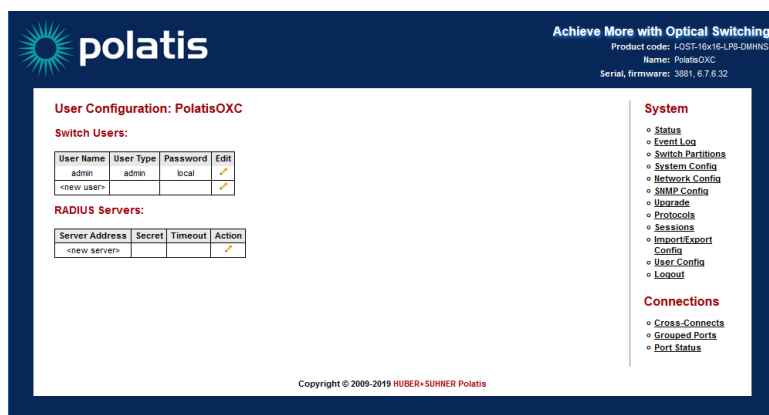
The **Import/Export** screen allows exporting the current switch settings and importing previously saved switch settings. Please consult the Web GUI User Manual for instructions.



7.3.13 User Config screen

The **User Config** screen allows creating new users with either administrative, basic user or view only permissions. Please consult the Web GUI User Manual for instructions.

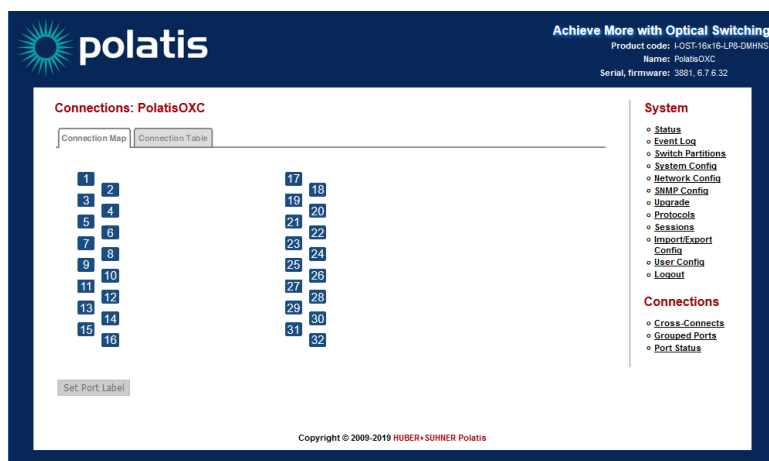
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7.3.14 Cross-Connects screen

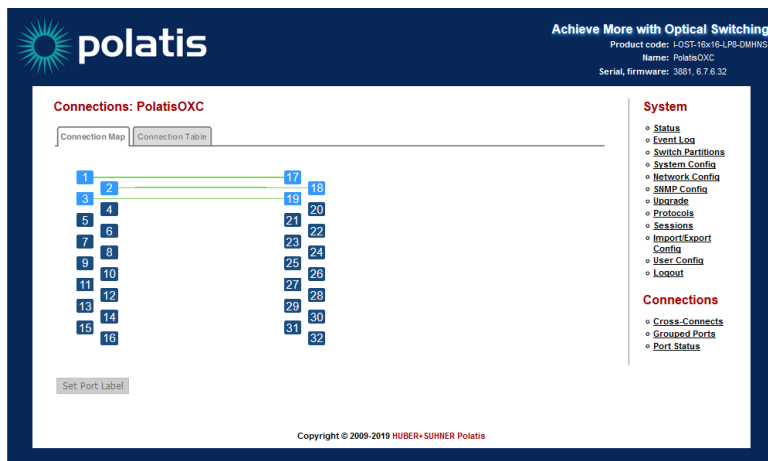
The **Cross-Connects** screen is the primary screen for making and changing switch optical connections. The illustrations below and following show an example of a 16x16 matrix switch.

The 16x16 matrix cross-connects screen shows input ports 1 through 16 on the left and output ports 17 through 32 on the right. Connections are simply made by selecting a port on the left, which when selected will turn amber, and then selecting the output port on the right you wish to connect to.

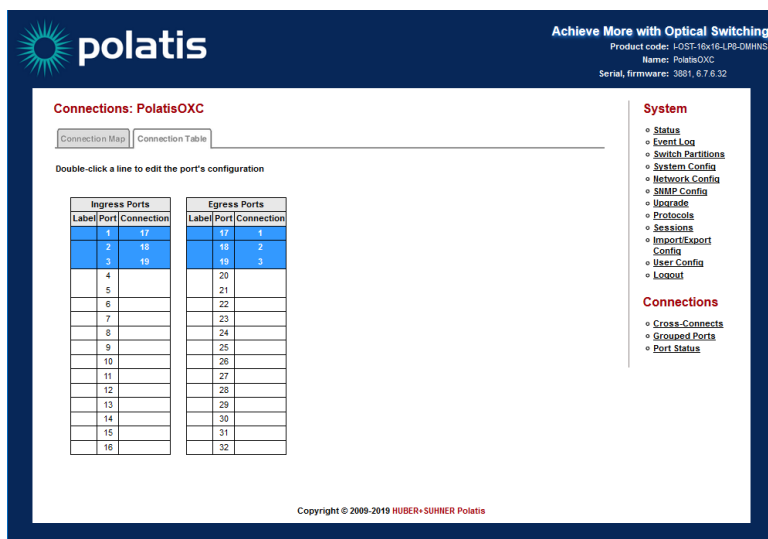


In the following example, three connections have been made between ports 1 and 17, 2 and 18 and 3 and 19. Active connections are shown by a green line.

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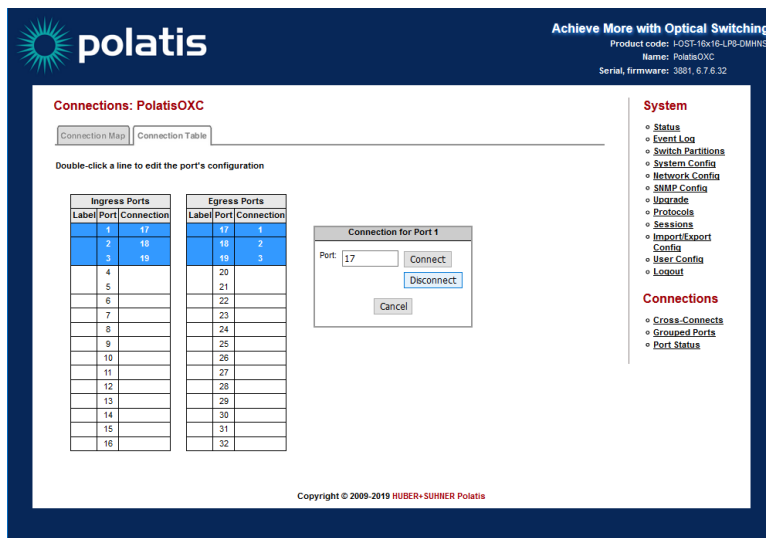
If desired, the cross-connects can be shown in a tabular format. This can be done by selecting the Connection Table tab at the top which shows the following screen:



Connections can be made by clicking on a connection box in the input port column and entering an output port number or clicking on a connection box in the output port column and entering an input port number.

Selecting the input and output ports on an already existing connection will delete that connection. Selecting either the input or output port on an existing connection and then selecting another port will disconnect the existing path and create a new connection.

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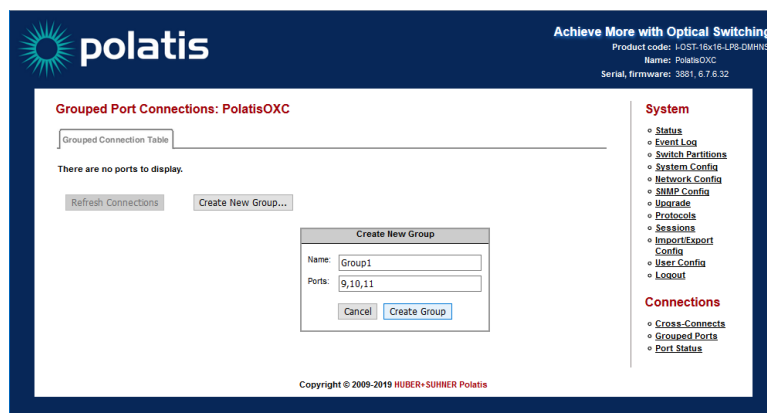


Select **Disconnect** and then confirm the disconnect by selecting **OK** in the pop-up to proceed with the change.

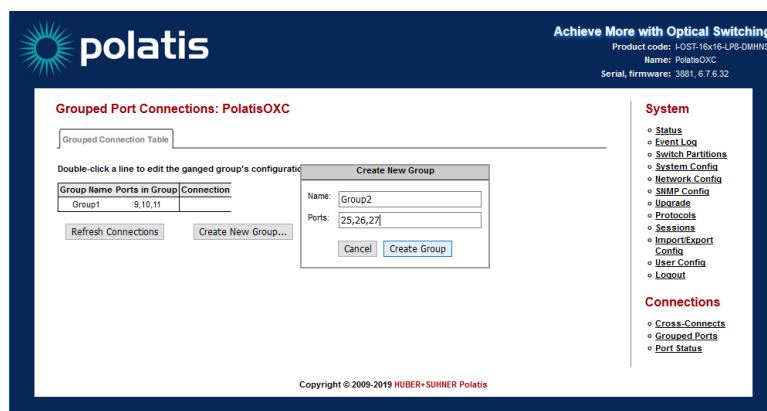
7.3.15 Grouped Ports screen

This allows a group of ports to be assigned and connected to another group of ports. The port groups must be defined with a quantity of import ports which matches the quantity of output ports in the desired connection group and vice-versa.

1. Create a group:

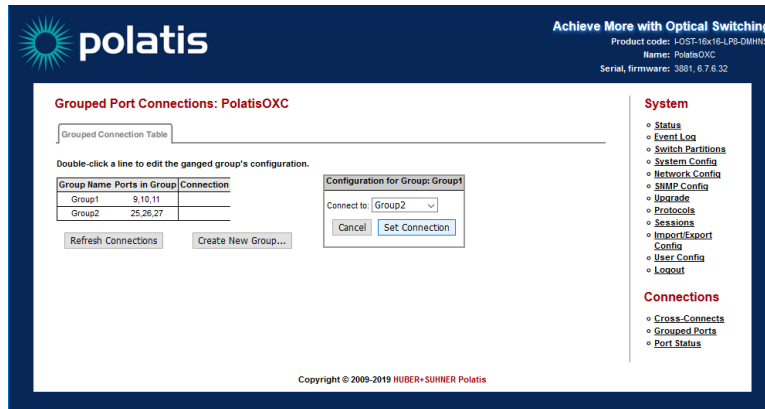


2. Next create another group.

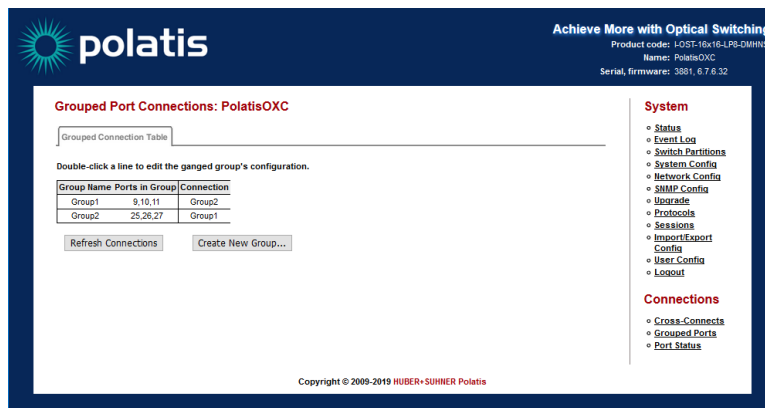


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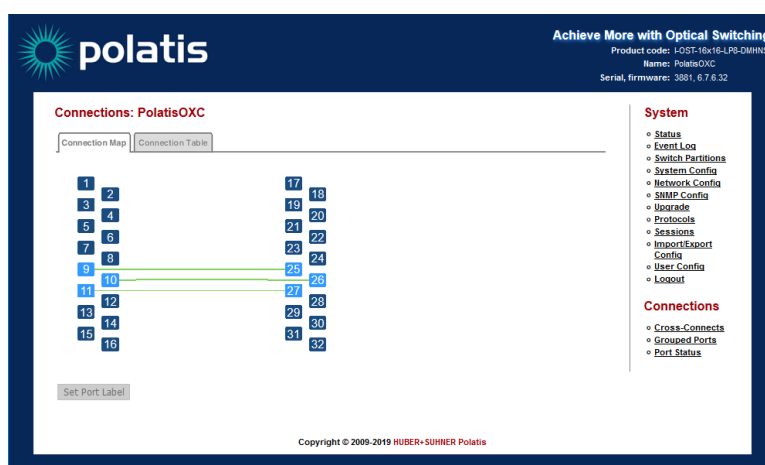
- Now connect the groups by double clicking in the **Connection** column to bring up the pop-up.
- Select the group to connect to and choose **Set Connection**.



- After selecting **Set Connection**, the groups are shown connected:



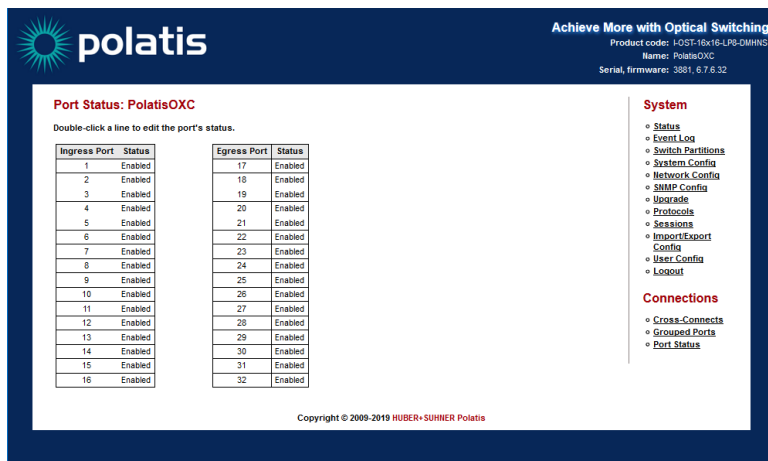
- The regular cross-connection screen will now show the three individual connections making up the group connection:



7.3.16 Port Status screen

The **Port Status** screen allows enabling or disabling each and every port. Normally all ports are enabled, but if you want to prevent one or more ports to make any connections, they can be disabled on this screen.

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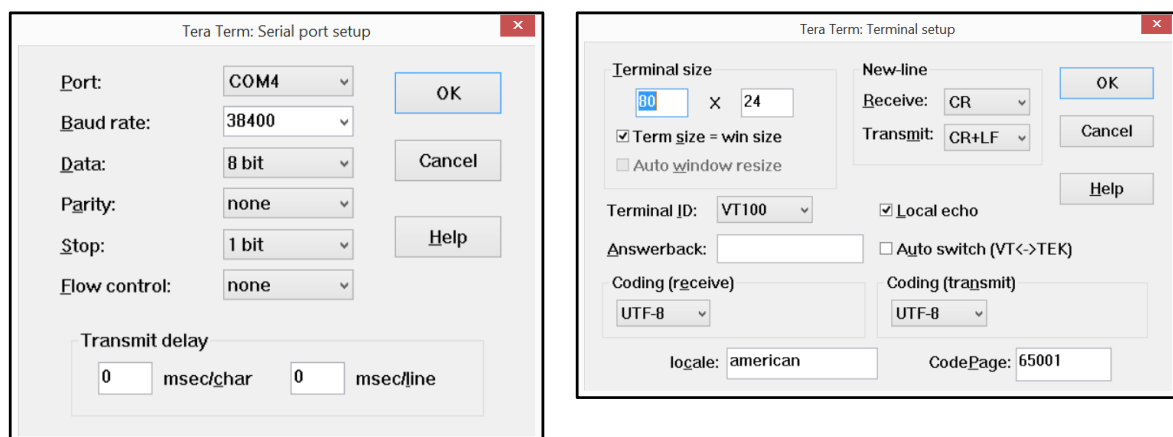
7.4 Other command interfaces

7.4.1 SCPI command interfaces

On most POLATIS switches, the industry standard raw SCPI command line interface is available on the serial port and Ethernet port 5025.

Please reference the SCPI User Manual for a full description of the SCPI command set.

The IP address can also be set by connecting to the SCPI interface on the serial port, or an existing Ethernet connection to port 5025. The serial port settings (shown for COM4 but set appropriately for your PC) are as follows, and you must turn on local echo.



7.4.1.1 Serial port connection settings

Once you have established a connection:

1. Verify the SCPI interface by issuing the following command to identify the switch:

Table 8 SCPI command to identify the switch

Command	*idn?
Example response	Polatis,I-OST-16x16-LP8-DMHNS,003881,6.7.6.32

2. Then issue the following command to verify the current IP settings:

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Table 9 SCPI command to verify the current IP settings

Command	:syst:comm:netw:addr?
Example response	IP="192.168.0.1" Gateway="0.0.0.0" Mask="255.0.0.0" Broadcast="192.168.0.255"

3. Then issue the command to enter the new IP settings in the same order they were just queried:

Table 10 SCPI command to enter new IP settings

Command	:syst:comm:netw:addr "10.10.1.102", "10.10.1.1", "255.0.0.0", "10.255.255.255"
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4. Verify the changed IP settings:

Table 11 SCPI command to verify IP settings changed

Command	:syst:comm:netw:addr?
Example response	IP="10.10.1.102" Gateway="10.10.1.1" Mask="255.0.0.0" Broadcast="10.255.255.255"

After verifying the new settings are correctly shown, power cycle the switch for the new settings to take effect. When the switch is power cycled, all optical connections are lost.

If it is necessary to change the IP settings without losing optical connections, please use the Web GUI. This can be done by making the changes on the Web GUI network config screen and selecting the check box "Check this box to reset the network card after configuring the network settings" and then selecting **Submit** will reset the IP address without affecting optical connections.

7.4.2 TL1 Command interfaces

The TL1 Command Line Interface is available on Ethernet port 3082. Please reference the TL1 User Manual for a full description of the TL1 command set.

7.4.3 SNMP interfaces

SNMP is available via the Ethernet interface. Please reference the SNMP User Manual for a full description of the SNMP command set.

7.4.4 NETCONF interfaces

NETCONF is available via the Ethernet interface. Please reference the NETCONF User Manual for a full description of the NETCONF command set.

7.4.5 RESTCONF interfaces

RESTCONF is available via the Ethernet interface. Please reference the RESTCONF User Manual for a full description of the RESTCONF command set.

8 Maintenance and operation

8.1 Fiber connection

All POLATIS switches are terminated with high performance optical connectors.

To connect the switch to other equipment, the appropriate single mode fiber patch cords should be used.

The POLATIS switch is a very high-performance fiber optic switch. To maintain this performance, it is important to observe the following fiber handling precautions.

8.1.1 Connecting and disconnecting

Patch cords should not carry significant optical powers when mating and de-mating with the switch optical bulkhead connectors, as this could damage the patch cord, the connectors or the switch and further cause an eye-hazardous situation.

It is therefore recommended that Class 1 laser limits be adhered to when connecting or disconnecting optical equipment to or from the switch.

8.2 Fiber cleaning

All patch cords to be used should be cleaned and inspected with an eye-safe fiber optic inspection scope prior to connection to the optical switch. If there are any visible defects or contamination on the connector, the connector should be re-cleaned and inspected again prior to connection to the switch.

8.3 Unused switch ports

Unused switch ports should always be covered with the bulkhead dustcover flaps or connector caps provided to prevent ingress of dirt onto the bulkhead connector.

8.4 Cleaning bulkhead adaptors

Cleaning of the optical switch bulkhead adaptors should not generally be necessary if the precautions listed above are observed. If an increased loss from a port to all other ports on the opposing side of the switch is observed, then it is likely that that port has become contaminated. In such cases the connector should be cleaned through the bulkhead adapter using an appropriate optical bulkhead cleaning tool.

 **Note:** Do not remove the chassis cover as this will invalidate the warranty.

8.5 Field replaceable units

8.5.1 Power supplies

The dual hot-swap power supplies can be fitted with either two Type H 100-240VAC (47-63Hz) autosensing power supplies, two Type T -48VDC (-40 to -72VDC) power supplies, or a combination of both. These units are all hot swappable.

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In all configurations, the switch is capable of operating on a single supply, and in the event of a failure, the failed unit can be replaced without affecting switch operation. It is suggested to remove the IEC-320 power cord from the Type H to be replaced, before removing the supply from the chassis. Similarly, it is suggested to disconnect the 48VDC external supply from the Type T PSU, before removing it from the chassis.



The HUBER+SUHNER Polatis part number 85135215 (P-1002-2382) is for the Type H, Zippy G1X-1060V12 100-240 VAC PSU.



The HUBER+SUHNER Polatis part number 85114741 (P-1002-2053) is for the Type T, Zippy DG1X-1060V12 -48VDC PSU.

8.5.2 Network Interface Controller

The Network Interface Controller (NIC) is field swappable should an issue arise. For the single NIC with dual Ethernet connections, the switch needs to be powered down before removing the NIC.

For the hot-swap NICs, either one may be removed without turning off the switch. If the active NIC is removed, the other NIC will transition to the active within 10 seconds, and its status LED will transition from flashing to steady green. If the standby NIC is removed, the active NIC will continue to operate the switch. The NIC or a replacement NIC can be re-inserted once the status LED on the operating NIC becomes steady green.

To remove a NIC:

1. Loosen the two knurled thumb screws until they disengage from their seating threads in the chassis.
2. Gently pull forward on both of them.
3. Remove the NIC from the chassis.

To replace a NIC:

1. Make sure the sides of the printed circuit board engage in the side slots.
2. Carefully seat the NIC fully.

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3. Tighten the two knurled thumb screws.

After the NIC is fully seated and the screw tightened, for the Dual Ethernet NIC, you may turn on the switch.

The switch settings for Ethernet address and other parameters will have to entered, just as when the switch was originally configured.

For the hot-swap NICs, as soon as the re-inserted NIC is engaged, it will begin a boot sequence and the status LED will transition to a flashing green, as it will assume the Standby status. The Ethernet settings will be maintained through this process, with the primary Ethernet Port 0 address available on the NIC in the lower slot, and the secondary Ethernet Port 1 address available on the NIC in the upper slot.

The POLATIS part number 85033989 (A-1000-10434) is for the Dual GbE Ethernet NIC Assy.



The POLATIS part number 85033746 (A-1000-9249) is for the Hot Swap GbE NIC Assy.



9 Switch connectivity check

The switch connectivity can be checked using either a visible fault locator or an appropriate loss measurement system, using the chosen control interface, to exercise the switch connection map. This will verify the optical path continuity through the switch and correct communication to the switch.

Please refer to the relevant interface operations manual for details of controlling the switch.

10 Software upgrade

HUBER+SUHNER Polatis customer support will provide notification of any required software upgrades. Requested feature enhancements may require an upgrade that often can be performed in the field. Please contact HUBER+SUHNER Polatis customer support with the model and serial number of your POLATIS switch.

11 HUBER+SUHNER Polatis customer support

Please have available the model and serial number of your POLATIS switch before you contact us:

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